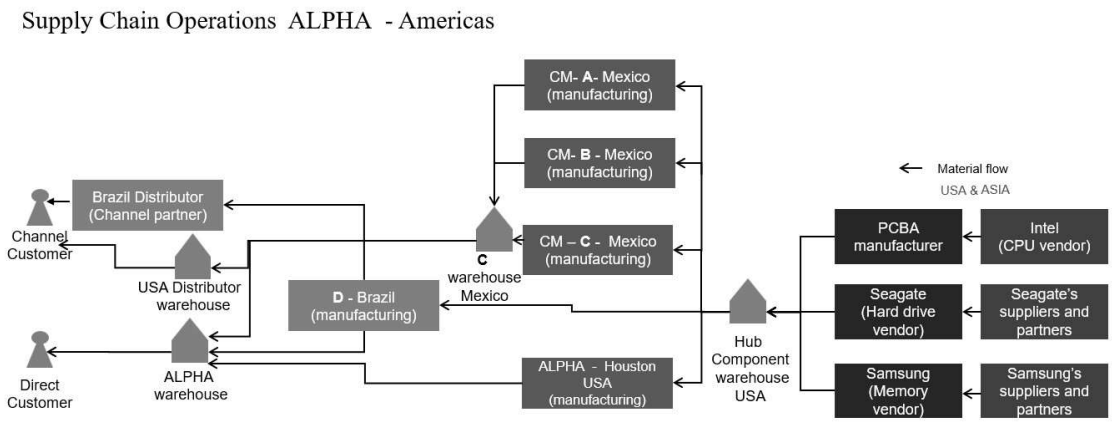
# APPENDIX A – ALPHA ORGANIZATION: SC Framework

Customer demand originates in the upper left, with the Brazil Distributor (Channel Partner), while overall planning starts from right to left. Components are supplied and processed in regional factories, and inventory is optimized based on order intake. Each SC node presents challenges, including misalignment, high turnover, accumulation of backlog, and excess inventory if plans are not synchronized. The Figure 6 presents an end-to-end view of Alpha's SC framework.




Legend

| Acronym | Full Term | Description |
|---|---|---|
| CM | Contract Manufaturer | Third party company responsable for assembly production |
| PCBA | Printed Circuit Board Assembly | Assembled circuit board with all required eletronics components |
| DC | Distribution Center | Facility for product storage and dispatch |
| WH | Warehouse | Storage facility used for inventory management |
| Vendor | vendor | Entity providing raw materials or componentes ( typically downstream ) |
| Supplier | supplier | Entity providing raw materials or componentes (tipically upstream) |

Figure 6 – Alpha organization SC framework
Source: own elaboration.

# APPENDIX A:

# A.1 Simulation vs. ANN Performance comparison

## Comparison of Simulation and ANN Performance Across Scenarios

Table 14 – Scenario I: Simulation vs. ANN Performance comparison

| Event | Metric | Simulation | ANN | Δ% | Key Insight |
|---|---|---|---|---|---|
| Free Disruption | Revenue | 1666.7 | 1698.7 | 1.9 | ANN captures more demand |
| | Profit | 685.8 | 667.3 | −2.7 | Higher costs reduce profit |
| | Total Cost | 980.9 | 991.0 | 1.0 | Marginal cost increase |
| | Service Level | 1.000 | 1.000 | 0.0 | Perfect SL maintained |
| | TTR (days) | 0 | 0 | 0.0 | Instant recovery |
| DC Miami Failure | Revenue | 1613.3 | 1615.8 | 0.2 | Minimal revenue impact |
| | Profit | 625.9 | 697.9 | +11.5 | ANN excels in contingency profit |
| | Total Cost | 987.3 | 917.8 | -7.0 | Cost efficiency during disruption |
| | Service Level | 0.950 | 0.902 | -5.1 | Worse customer fulfillment |
| | TTR (days) | 15 | 14 | -6.7 | Slightly faster recovery |
| DC Brazil Failure | Revenue | 1630.1 | 1611.1 | −1.2 | Minor demand loss |
| | Profit | 630.9 | 697.9 | +10.6 | Better contingency planning |
| | Total Cost | 999.2 | 913.1 | -8.6 | Improved cost control |
| | Service Level | 0.980 | 0.901 | -8.1 | Struggles with demand swings |
| | TTR (days) | 10 | 9 | -10.0 | Faster resolution |
| Demand Variation | Revenue | 1708.4 | 1609.5 | −5.8 | Demand volatility impact |
| | Profit | 684.2 | 696.2 | 1.7 | Better profit retention |
| | Total Cost | 1024.2 | 913.3 | -10.8 | Significant cost savings |
| | Service Level | 0.980 | 0.901 | -8.1 | Service quality decline |
| | TTR (days) | 0 | 0 | 0.0 | No recovery needed |
| Factory Mexico Failure | Revenue | 1666.7 | 2737.8 | +64.3 | Major revenue surge |
| | Profit | 629.0 | 695.9 | +10.6 | Strong profit recovery |
| | Total Cost | 1037.7 | 2041.9 | +96.8 | Substantial cost increase |
| | Service Level | 1.000 | 1.000 | 0.0 | Uninterrupted service |
| | TTR (days) | 0 | 0 | 0.0 | Immediate restoration |
| Factory Brazil Failure | Revenue | 1509.4 | 1511.4 | 0.1 | Stable revenue |
| | Profit | 665.9 | 668.0 | 0.3 | Marginal improvement |
| | Total Cost | 843.5 | 843.4 | 0.0 | Identical costs |
| | Service Level | 0.910 | 0.911 | 0.1 | Slightly better SL |
| | TTR (days) | 0 | 0 | 0.0 | No downtime |

**Note:** Positive Δ% (green) indicates ANN improvement; negative Δ% (red) shows degradation. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.

Table 15 – Scenario II: Simulation vs. ANN Performance comparison

| Event | Metric | Simulation | ANN | Δ% | Key Insight |
|---|---|---|---|---|---|
| Free Disruption | Revenue | 3333.4 | 3333.4 | 0.0 | No baseline change |
| | Profit | 1153.3 | 1153.3 | 0.0 | Identical performance |
| | Total Cost | 2180.2 | 2180.2 | 0.0 | No cost variance |
| | Service Level | 1.000 | 1.000 | 0.0 | Perfect SL |
| | TTR (days) | 0 | 0 | 0.0 | Zero downtime |
| DC Miami Failure | Revenue | 3281.5 | 3288.2 | 0.2 | Minimal impact |
| | Profit | 1085.7 | 1058.9 | -2.5 | Short-term profit dip |
| | Total Cost | 2195.8 | 2229.3 | 1.5 | Higher mitigation costs |
| | Service Level | 0.988 | 0.901 | -8.8 | Significant SL drop |
| | TTR (days) | 100 | 0 | -100.0 | ANN eliminates downtime |
| DC Brazil Failure | Revenue | 3333.4 | 3439.4 | +3.2 | Post-disruption demand boost |
| | Profit | 1001.7 | 1029.1 | 2.7 | Better profit retention |
| | Total Cost | 2331.8 | 2410.3 | 3.4 | Higher operational costs |
| | Service Level | 1.000 | 1.000 | 0.0 | Full service maintained |
| | TTR (days) | 0 | 0 | 0.0 | No recovery time |
| Demand Variation | Revenue | 3457.4 | 3385.2 | −2.1 | Demand fluctuation |
| | Profit | 1157.1 | 1097.5 | -5.2 | Profit sensitivity |
| | Total Cost | 2300.3 | 2287.7 | −0.6 | Minor cost savings |
| | Service Level | 1.000 | 1.000 | 0.0 | Stable service |
| | TTR (days) | 0 | 0 | 0.0 | No disruption |
| Factory Mexico Failure | Revenue | 3292.2 | 3292.2 | 0.0 | No change |
| | Profit | 1088.5 | 1088.5 | 0.0 | Identical results |
| | Total Cost | 2203.7 | 2203.7 | 0.0 | Cost neutral |
| | Service Level | 0.988 | 0.988 | 0.0 | Consistent performance |
| | TTR (days) | 50 | 50 | 0.0 | Fixed recovery period |
| Factory Brazil Failure | Revenue | 1880.1 | 1898.1 | 1.0 | Slight improvement |
| | Profit | 769.8 | 1081.5 | +40.5 | Major profit recovery |
| | Total Cost | 1110.3 | 816.6 | -26.5 | Significant cost reduction |
| | Service Level | 1.000 | 1.000 | 0.0 | Unaffected service |
| | TTR (days) | 0 | 0 | 0.0 | Instant restoration |

**Note:** Positive Δ% (green) indicates ANN improvement; negative Δ% (red) shows degradation. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.

Table 16 – Scenario III: Simulation vs. ANN Performance comparison

| Event | Metric | Sim. | ANN | Δ% | Key Insight |
|---|---|---|---|---|---|
| Free Disruption | Revenue | 2135.8 | 2137.8 | 0.1 | Stable baseline |
| | Profit | 671.5 | 679.5 | 1.2 | Slight improvement |
| | Total Cost | 1464.3 | 1458.3 | −0.4 | Minor cost savings |
| | Service Level | 1.000 | 1.000 | 0.0 | Perfect service |
| | TTR (days) | 0 | 0 | 0.0 | No disruption |
| DC Miami Failure | Revenue | 2143.1 | 2145.1 | 0.1 | Resilient revenue |
| | Profit | 175.8 | 740.5 | +321.1 | Extraordinary recovery |
| | Total Cost | 1967.3 | 1404.6 | -28.6 | Dramatic cost reduction |
| | Service Level | 0.980 | 0.901 | -8.1 | Service degradation |
| | TTR (days) | 15 | 14 | -6.7 | Faster resolution |
| DC Brazil Failure | Revenue | 2139.1 | 2121.3 | −0.8 | Minor revenue dip |
| | Profit | 788.2 | 760.5 | -3.5 | Profit sensitivity |
| | Total Cost | 1351.0 | 1360.8 | 0.7 | Comparable costs |
| | Service Level | 1.000 | 1.000 | 0.0 | Uninterrupted service |
| | TTR (days) | 17 | 16 | -5.9 | Quicker recovery |
| Demand Variation | Revenue | 2199.0 | 2118.2 | −3.7 | Demand volatility |
| | Profit | 1058.7 | 1087.7 | 2.7 | Profit resilience |
| | Total Cost | 1140.3 | 1030.5 | -9.6 | Cost efficiency |
| | Service Level | 1.000 | 1.000 | 0.0 | Stable performance |
| | TTR (days) | 16 | 15 | -6.3 | Faster stabilization |
| Factory Mexico Failure | Revenue | 2143.1 | 2145.1 | 0.1 | Revenue stability |
| | Profit | 980.2 | 1088.7 | +11.1 | Strong profit growth |
| | Total Cost | 1162.9 | 1056.4 | -9.2 | Improved cost control |
| | Service Level | 1.000 | 1.000 | 0.0 | Consistent service |
| | TTR (days) | 15 | 14 | -6.7 | Quicker restoration |
| Factory Brazil Failure | Revenue | 2143.1 | 2156.4 | 0.6 | Revenue uptick |
| | Profit | 1179.5 | 1181.4 | 0.2 | Marginal improvement |
| | Total Cost | 963.6 | 975.0 | 1.2 | Slight cost increase |
| | Service Level | 1.000 | 1.000 | 0.0 | Perfect reliability |
| | TTR (days) | 17 | 16 | -5.9 | Faster crisis resolution |

Source: own elaboration.

**Note:** Positive Δ% (green) indicates ANN improvement; negative Δ% (red) shows degradation. TTR = Time to Recover, SL = Service Level.

Source: own elaboration.